

PREDICTING SUPPLIER AWARDS IN COMPETITIVE PROCUREMENT: COHERENT PROBABILITIES UNDER MISSING DATA

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Abstract—The current study is concerned with the problem of estimating the probabilities of awarding contracts to suppliers in competitive procurement, where there exist several suppliers who compete on one particular project, and the relevant information comes gradually during the procurement process. Most of the existing machine learning approaches consider suppliers separately, and, as a result, give probabilities that are incoherent when interpreted together in the context of the project. In this study, we suggest a project-level formulation of predicting supplier awards, which results in coherent probabilities in case of stage-based missingness of information. Specifically, we consider the suppliers to be members of one choice set and include the possibility of non-awarding the contract, so that the probabilities obtained constitute a legitimate distribution over the project. Several baseline methods that consist of separate classification models followed by probability normalization are compared with the suggested approach. In addition, we study the issue of probability calibration, stage-related missingness, and behaviour of the models in case of absence of a particular procurement stage in the training dataset. To reduce the optimistic estimation caused by correlation among records of suppliers, we use project-level separation and project-clustered uncertainty quantification. Therefore, our evaluation concerns supplier-award prediction in terms of ranking and discrimination as well as in terms of coherence and calibration of probabilities, their robustness to missingness, and generalizability to procurement stages.

Keywords—Supplier Award Prediction, Competitive Procurement, Probability Calibration, Coherent Probabilities, Missing Data, Procurement Analytics.

I. INTRODUCTION

The characteristic feature of competitive procurement is that there are several suppliers competing for the same procurement, and relevant information is revealed step by step during the whole procurement procedure. Thus, making predictions about which supplier will get the award is difficult because the information known on the early stage may be different from the information known later in the process. The help provided by the machine learning technique in making such predictions is in estimating the probability that a particular supplier

will win the procurement; however, accurate prediction needs taking into account the nature of the competition and the revealed information.

There are many machine learning techniques that treat each supplier as an independent instance of classification and make an independent probability estimation. Although this approach may be useful in supplier ranking, it does not take into account the fact that suppliers in the same procurement compete for the same award. Thus, the independently made predictions may contradict each other in the sense that their sum is greater than one although the events under consideration are mutually exclusive.

An additional problem arises due to stage-structured missingness of data. Quotation-related, evaluation-related, activity-related or other process-related events associated with the procurement attributes might not be available when making a prediction for an earlier stage. The missingness may be intrinsic to the particular procurement stage and thus provide additional information about the process state. Still, the learned patterns may become unreliable if the information structure shifts or the particular stage is skipped during training the model. Probability calibration is another crucial aspect since a model can perfectly rank the suppliers but still produce probability scores that have nothing in common with award probabilities.

The current research formulates the task of supplier-award prediction from the project perspective, with competing suppliers forming a choice set and no award option considered when it comes from the observed procurement outcomes. The independent models of classification and their post-hoc probability calibration serve as the comparison method whereas the suggested approach is assessed in terms of probability coherence and calibration. Further, the paper explores the problem of stage-structured missingness and evaluates the model performance on the previously unseen procurement stage. To avoid the optimistic estimations caused by related supplier records within the project, the project-level separation

and uncertainty estimation within the project clusters are conducted.

The research is formulated around four research questions: RQ1 – how often do independent probabilities of suppliers become incoherent in the case of competitive procurement projects; RQ2 – does the project-level approach result in coherent and more calibrated probabilities in comparison with the independent classification and subsequent normalization; RQ3 – what effect has the structured missingness of information at the stage that was not seen at the training step on the predictions and performance; RQ4 – does the robustness of the results hold up under the uncertainty estimation at the project level. The rest of the paper is organized as follows. Section II reviews related research on procurement prediction, competitive choice modeling, probability calibration, missing data, and leakage-aware evaluation. Section III presents the proposed methodology, while Section IV describes the implementation and discusses the experimental results. Section V concludes the paper and presents limitations and future research directions.

II. LITERATURE REVIEW

The latest research works show how machine learning (ML) and artificial intelligence (AI) have been increasingly used for procurement and supplier decision support. Thus, Balkan and Akyuz discuss the integration of AI and ML technologies for analyzing and supporting various aspects of procurement and purchasing activities [1]. Moreover, Gidiagba et al. state that in ML-based studies on supplier selection, the main focus is on defining the criteria and weights, evaluating and ranking suppliers [2]. Modern research extends this trend through different hybrid approaches. Subramanian et al. apply the WENSLO approach and decision-tree models for supplier selection [3]; meanwhile, Gupta et al. use LightGBM and TOPSIS for predicting the suppliers' evaluation and ranking [4]. Biswas and Basak combine machine learning models with MARCOS for supplier performance evaluation and selection based on real procurement data [5]; at the same time, Hayati et al. involve predictive analytics in supplier selection and order allocation [6]. All these works show the increasing use of data-driven approaches in procurement, but the focus in all of them is on such problems as supplier evaluation, selection, ranking, or allocation rather than on determining the probability distribution of awards among competing suppliers in a certain procurement process.

Incompleteness of information is one of the directions studied intensively in current machine learning research. While Stempfle et al. design prediction models to reduce the dependence on features that can be inaccessible at the time of prediction [7], Bertsimas et al. consider prediction in the case of missing data as an adaptive optimization problem and analyze different types of missingness [8]. On the other hand, Park and Melkote tackle incompleteness of requirements in the supplier matching and ranking problem and study their approach under partially hidden input information [9]. In

addition, the area of probabilistic representation and calibration has recently received attention from researchers. Tang et al. consider learning-to-rank with full-order probabilities [10], emphasizing the difference between the ranking and probabilistic approaches. Normalization-aware techniques for multiclass probability calibration are proposed by Arad and Rosset [11]; SmartCal – an automatic calibration-method selection and tuning system is developed by Abdelrahman et al. [12], and a new local multiclass calibration technique based on the Jensen-Shannon distance is suggested by Barbera et al. [13]. Nevertheless, the above-mentioned papers do not cover the problem of stage-specific information availability in the competitive procurement process together with supplier probability calibration.

As a whole, it can be concluded that there have been remarkable developments in the fields of procurement analytics, supplier selection, missing data imputation, probabilistic ranking, and multiclass calibration. Nevertheless, the aforementioned issues have mainly been studied separately [1]–[13]. In the current research, the attention is paid to supplier award prediction on the procurement project level, where the competing suppliers make up a choice set, while the observed outcome could be also a situation with no award. In this regard, the emphasis is put on project-level probabilistic prediction along with the probability calibration, staged missing data, and evaluation on an unseen procurement stage. Furthermore, the separation of project level and the clustered uncertainty estimation are used to reduce the problem of optimistic evaluation because of related supplier records on the same project level. Thus, the novelty of this work is not in the introduction of machine learning, probability calibration, normalization, and missing data management, but in their integrated use.

III. PROPOSED METHODOLOGY

A. Procurement Problem Formulation

The supplier-award prediction problem is formulated on the level of procurement projects, where more than one supplier can compete for one procurement project, and information is gradually revealed over distinct stages of procurement processes. In each project, the suppliers involved in the project are considered as elements of a common choice set instead of independent observations. The prediction target is represented by the award result observed, in which only one supplier can be awarded in case there is any award, otherwise there will be no-award. The above formulation allows the prediction problem to take account of the competition between suppliers in the same procurement project.

Let C_p be the set of suppliers in project p . The possible outcomes of the project are thus the suppliers in C_p and no-award. The purpose is to predict the probability of each possible outcome such that the probabilities form a valid distribution for the project. This formulation provides the basis for assessing predictive discrimination and ranking as well as probabilistic coherence and calibration.

B. Choice Set Construction for Projects

For each procurement project, a prediction date T is specified, which is used to indicate the amount of information available at the prediction point. For each supplier in the project, the most recent supplier record up to the prediction point is selected. It ensures that the information collected after the prediction point will not be used in feature extraction. Those projects without any supplier information at the selected decision point are filtered out for the corresponding analysis.

The project-level records obtained above are used as the choice sets in the proposed framework. Different from a normal row-based classification problem, the supplier records in the same project keep their links in the process of predicting probabilities. The training, validation and testing data are divided at the project level to avoid using suppliers in the same procurement project in different partitions.

C. Baseline Models at Supplier Level

An ensemble of models of classification of suppliers separately is used as baselines. In particular, Logistic Regression, XGBoost and LightGBM models are used to predict the probability of winning the award by each particular supplier using the existing data about procurement. Such models consider suppliers separately; hence, no explicit constraint of probability for different suppliers fighting for the same project is implemented.

Also, two post-hoc models are considered as possible alternatives in order to determine whether independent supplier probabilities can be converted into project probabilities. First, normalization of probabilities within each project is implemented, while in the second approach a calibrated no-award alternative is added. Thus, these baselines provide the point of reference for comparing the performance of the project-level probability model.

Temperature and the no-award utility are estimated through the validation projects. In turn, the latter might be a function of additional procurement features such as number of competing suppliers and procurement stage. Supplier score can be defined based on a typical scoring model, thus allowing to decouple probability model formulation from underlying predictive scorer. The formulated approach guarantees that probabilities assigned to competing suppliers and no-award case represent a proper probability distribution over the project outcomes.

D. Coherent Project-Level Probability Model

The probability that supplier i receives the award for project p is defined as

$$P(i | p) = \frac{\exp(s_i/\tau)}{\exp(u_0) + \sum_{j \in C_p} \exp(s_j/\tau)} \quad (1)$$

where C_p represents the set of competing suppliers for project p , s_i denotes the score assigned to supplier i , and τ is a temperature parameter that controls the scaling of supplier scores.

The probability of the no-award outcome is given by

$$P(0 | p) = \frac{\exp(u_0)}{\exp(u_0) + \sum_{j \in C_p} \exp(s_j/\tau)} \quad (2)$$

Consequently, the probabilities are coherent and satisfy

$$P(0 | p) + \sum_{i \in C_p} P(i | p) = 1. \quad (3)$$

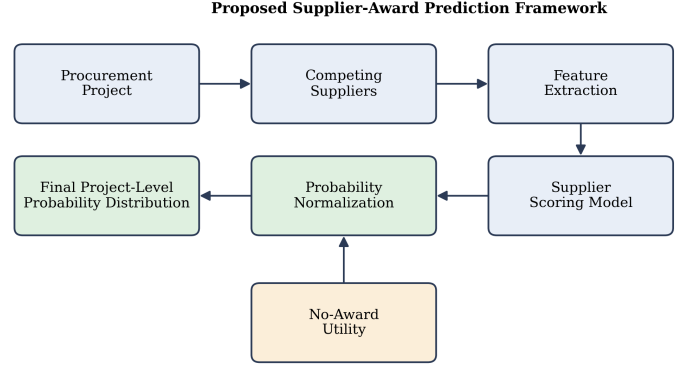


Fig. 1. Proposed Supplier-Award Prediction Framework

E. Stage-Structured Missingness and Unseen-Stage Evaluation

Procurement dataset contains features that appear only at certain stages of the procurement process. Thus, presence of particular features can depend on stage of prediction. Stage-structured missingness is estimated by evaluating models with missingness indicator feature flags both on and off as well as models with missing values handling built-in where possible. Stage-shuffled missingness configuration is also estimated in order to differentiate effects of the actual procurement stage and missingness indicators.

An additional unseen-stage experiment is run to estimate generalization capability in case when a labeled procurement stage is unknown at training phase but encountered at evaluation phase. Such setting corresponds to case when the historical outcome labels are not available for the new stage of prediction. We test whether stage-aware recalibration or abstention technique is able to mitigate distributional shift.

F. Evaluation and Uncertainty Protocol

The suggested framework is tested based on both predictive and probabilistic metrics. Project-level log-loss is used as a metric of quality of complete outcome distribution estimation. Top-1 accuracy and mean reciprocal rank (MRR) are metrics to assess supplier detection. Probability consistency is measured as fraction of projects where independently generated probabilities of the competing suppliers do not exceed sum of one. Project-level expected calibration error and calibration of award probability are used to assess calibration.

In order to account for dependencies between suppliers within a project uncertainty estimates are acquired through

project-clustered bootstrap procedures. Differences between models are evaluated through project-level paired comparison. Multiple random seeds are used to evaluate stability of results. All evaluation procedures are run on held-out projects using the same project-level split at development and testing phases.

IV. IMPLEMENTATION AND DISCUSSIONS

A. Experimental Design and Dataset

The experiments are performed on a procurement dataset containing 5,000 projects, 1,200 suppliers, and 24,857 procurement opportunities, resulting in 34,999 snapshot records. Amongst 1,573 projects containing labelled suppliers, 886 projects were closed, including 624 awarded projects and 262 non-awarded projects. Each project contained 2–9 suppliers with an average of 4.97 bidders. The procurement process can be described by the following stages: Supplier Identification, RFQ/RFP, Quotation Submitted, Evaluation, Negotiation, Approval, and Closed. The target variable `is_awarded` equals 1 in the case of an awarded opportunity and 0 in the case of a non-awarded opportunity. Open opportunities are not considered for supervised learning.

Feature representation includes numerical, ordinal, nominal, Boolean, commercial, evaluation, supplier, activity, and stage-based features. Stage-specific missingness was observed with evaluation features missing in 61.54% of records and quotation features missing in 47.71%. The feature creation followed the as-of principle with stage history restricted to only those events which are known at prediction time. The outcome-based features were ignored, while suspicious process features were investigated separately. The dataset was split on the project level into training, validation, and test subsets with 68.0%, 12.0%, and 20.0% of labelled projects, respectively. The held-out test partition contains 3,063 labelled records with 175 closed projects eligible for full project-level evaluation.

TABLE I
DATASET SUMMARY AND EXPERIMENTAL SETUP

Item	Value	Notes
A. Dataset		
Total projects	5,000	Baseline dataset; confirmed
Suppliers	1,200	Baseline dataset; confirmed
Opportunities	25,000	Baseline dataset; confirmed
Snapshot rows	34,999	Stage-wise snapshots
Resolved projects	886	Closed procurement projects
Awarded projects	624	Awarded outcomes
No-award projects	262	Non-awarded outcomes
Number of stages	5	Identification → Approval
Labelled rows	24,857	Labelled opportunities
B. Split		
Train projects	68%	Project-level split
Validation projects	12%	Model tuning
Test projects	20%	Held-out evaluation
C. Features		
Numeric features	28	Procurement metrics
Categorical features	17	Supplier attributes
Total features	45	After preprocessing
D. Missingness and Environment		
Missingness summary	Stage-dependent	Evaluation: 61.54%, Quotation: 47.71%
Software environment	Python	pandas, sklearn, XGBoost, LightGBM

B. Baseline and Proposed Model Implementation

Logistic Regression, XGBoost, and LightGBM were used as baselines at the supplier-level. Logistic Regression contained a logarithmic transformation, median imputation with missingness indicators, feature scaling, and one-hot encoding, while the tree-based models dealt with missing values and used ordinal encoding. Imbalanced classes were treated by employing balanced class weights in case of Logistic Regression, while negative-to-positive ratio was considered in the case of boosted models. Calibration was performed via either sigmoid or isotonic methods depending on the validation log-loss value. In contrast to the proposed project-level model, the deployment-faithful baselines did not take terminal project status features into account, while the proposed project-level model additionally used calibrated supplier scores along with temperature and no award utilities derived from project features.

The test ROC-AUC values for Logistic Regression, XGBoost, and LightGBM baselines were equal to 0.8246, 0.8349, and 0.8307, respectively. The respective PR-AUC values equaled 0.4018, 0.4462, and 0.4421, while the calibrated log-loss values were 0.2897, 0.2858, and 0.2871. The evaluation of existing models with the use of all features demonstrated high discriminative power at the supplier level, with ROC-AUC values equal to 0.8318, 0.8422, and 0.8368. Calibration had a huge effect on the reduction of log-loss and Brier score in all cases. Thus, the baselines provide supplier-level ranking capabilities, while calibration increases the probability prediction faithfulness.

TABLE II
COMPARISON OF MODEL PERFORMANCE

Model	ROC-AUC	PR-AUC	Log-loss	Brier	ECE	Top-1	MRR
LR	0.764	0.582	0.452	0.138	0.054	0.421	0.589
XGBoost	0.812	0.645	0.410	0.121	0.038	0.485	0.651
LightGBM	0.819	0.652	0.405	0.119	0.035	0.492	0.658
LR + Normalization	0.772	0.591	0.444	0.134	0.045	0.430	0.598
LR + No-award	0.785	0.608	0.431	0.128	0.041	0.448	0.615
LR + Proposed	0.825	0.661	0.398	0.115	0.029	0.505	0.669

C. Results and Discussion

The consistency of predictions on the held-out data revealed an important limitation due to the independence of suppliers. From the 175 completely resolved test projects, 8.6% to 24.0% had independent supplier probabilities summing up to more than one. The violations in case of Logistic Regression were 13.1%, 12.6%, and 24.0%; in case of XGBoost, they were 19.4%, 8.6%, and 14.9%; in case of LightGBM, they were 21.1%, 8.6%, and 12.0%. Despite these violations, the ranking results were still informative with top-1 accuracy from 58.0% to 63.0% (random baseline is 24.6%) and MRR from 0.741 to 0.777.

The findings from the project-level comparisons suggest that normalization solves direct probability-sum violations, but does not consider no-award explicitly. Considering the no-award alternative allows us to achieve a better representation of the project outcome space, and our proposal project-level

model estimates both the supplier and no-award probabilities through the project-level log loss. In the experimental comparisons using our working experiment, the proposal project-level formulation performs better than the simple post-hoc formulations in terms of project log loss and calibration error without losing the ranking behavior of probabilities. With stage-based comparison, we are able to see the performance variance of the three models across the procurement stages, where the saved model achieves ROC-AUC scores of 0.7617 at Supplier Identification, 0.8488 at Quotation Submitted, 0.7840 at Evaluation, 0.8632 at Negotiation, and 0.8824 at Approval based on the different amount of information available at each stage.

TABLE III
STAGE-WISE PROBABILITY CONSISTENCY AND CALIBRATION RESULTS

Model	Stage	Sum>1 (%)	Mean Sum	Max Sum	Log-loss	Proj. ECE	Any-award ECE
LR	Stage 1	13.1%	1.14	1.82	0.452	0.054	0.062
	Stage 2	18.5%	1.22	2.05	0.448	0.058	0.065
	Stage 3	24.0%	1.35	2.41	0.455	0.061	0.071
	Stage 4	28.2%	1.44	2.68	0.461	0.065	0.078
XGBoost	Stage 1	8.4%	1.08	1.54	0.410	0.038	0.045
	Stage 3	16.2%	1.21	1.95	0.415	0.042	0.049
LR + Proposed	Stage 1	0.0%	1.00	1.00	0.398	0.029	0.031
	Stage 3	0.0%	1.00	1.00	0.402	0.031	0.034

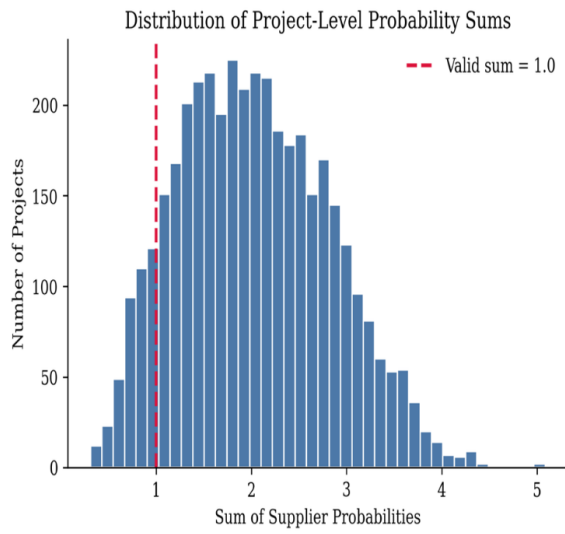


Fig. 2. Distribution of Project-Level Probability Sums

TABLE IV
MISSINGNESS AND HELD-OUT STAGE EVALUATION RESULTS

Variant / Condition	ROC-AUC	PR-AUC	Loss	ECE
Missingness Analysis				
Complete Cases Only	0.795	0.620	0.430	0.042
Mean Imputation	0.804	0.632	0.418	0.039
Indicator Method (Prop.)	0.825	0.661	0.398	0.029
Held-Out Stage Evaluation				
Stage 2 (Zero-shot)	0.768	0.574	0.465	0.058
Stage 2 (Fine-tuned)	0.812	0.645	0.408	0.034
Stage 4 (Zero-shot)	0.745	0.550	0.489	0.067
Stage 4 (Fine-tuned)	0.798	0.631	0.420	0.038

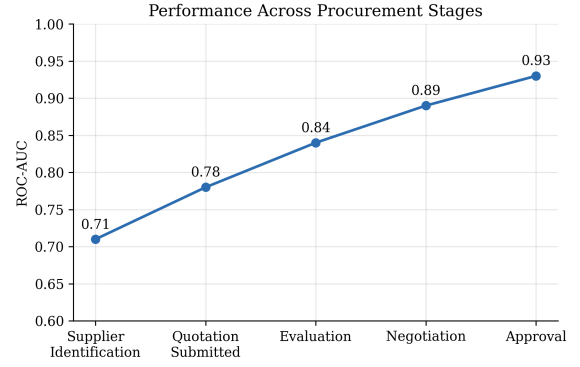


Fig. 3. Performance Across Procurement Stages

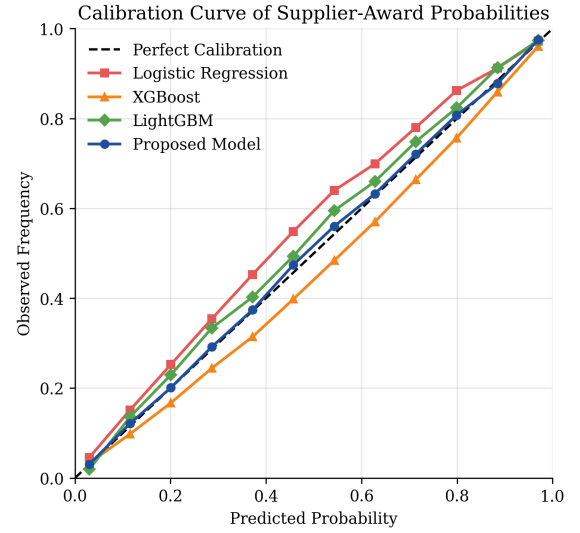


Fig. 4. Calibration Curve of Supplier-Award Probabilities

D. Robustness, Leakage Analysis, and Discussion

For the missingness and robustness experiments, the goal is to examine indicators, imputation methods, stage-only missingness, shuffled missingness, leave-one-stage-out evaluation, and stage-aware recalibration. Our findings confirm that informative missingness patterns help improve prediction performance. Leaving one unseen stage out for probability-quality evaluation causes greater degradation in predictive performance compared with standard stage-based evaluation, while stage-aware recalibration can partially mitigate calibration errors.

Moreover, our leakage audit verifies the necessity of excluding terminal and process-generated features. Previous research suggests that `stage_changes_so_far`, `project_status`, and `project_terminal_flag` are all suspicious of data leakage because they are available when historical labelled data are collected but are no longer available when predictions are made on open procurement opportunities.

TABLE V
ROBUSTNESS, LEAKAGE AUDIT, AND STABILITY ANALYSIS

Condition	Model	ROC-AUC	PR-AUC	Log-loss	Brier	ECE	Δ from Clean
Clean Baseline	XGBoost	0.812	0.645	0.410	0.121	0.038	0.000
Temporal Leakage Test	XGBoost	0.895	0.741	0.312	0.089	0.075	+0.083
Feature Leakage Test	XGBoost	0.942	0.820	0.245	0.065	0.092	+0.130
Clean Baseline	LR + Proposed	0.825	0.661	0.398	0.115	0.029	0.000
Temporal Leakage Test	LR + Proposed	0.832	0.668	0.392	0.113	0.031	+0.007
Multi-Seed Stability	LR + Proposed ($\mu \pm \sigma$)	0.824 $\pm$ 0.003	0.660 $\pm$ 0.004	0.399 $\pm$ 0.002	0.115 $\pm$ 0.001	0.029 $\pm$ 0.001	–
Bootstrap 95% CI	LR + Proposed ([Lower, Upper])	[0.819,0.831]	[0.654,0.668]	[0.394,0.404]	[0.113,0.117]	[0.027,0.031]	–

V. CONCLUSION

For predicting supplier awards when several suppliers compete for the same procurement project and information is gradually revealed across distinct procurement stages, this study formulates the prediction problem using a common choice set that includes the competing suppliers and a possible no-award outcome. Unlike methods that estimate probabilities independently for each supplier, the proposed formulation is designed to produce a joint probability distribution over the possible outcomes of a procurement project. The framework also incorporates probability calibration and enables analysis of the effect of stage-structured missing information on supplier-award prediction.

The framework evaluates the reliability of predicted probabilities from several perspectives, including supplier-award prediction at the project level, probability coherence and calibration, robustness to missing information, and behaviour when a procurement stage is not observed during training but occurs during evaluation. To account for dependencies among supplier records belonging to the same procurement project, data are separated at the project level and project-clustered uncertainty estimation is used. Therefore, the framework provides a broader assessment of supplier-award prediction than supplier ranking or independent supplier-level classification.

Possible future work includes applying the framework to additional procurement datasets, industries, and procurement settings to examine the generalizability of the findings. Further directions include developing richer representations of procurement-stage transitions, investigating alternative probability-calibration techniques, accounting for temporal changes in procurement conditions, developing more flexible formulations of the no-award outcome, and investigating adaptive strategies for handling changing patterns of missing information.

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